Nesryne Mejri

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Research interests: Deepfake detection, Unsupervised Anomaly detection, Unsupervised domain adaptation

Employment

Aug. 2025 – Present — **Postdoctoral Researcher** – Computer Vision, Imaging and Machine Intelligence (CVI^2) Research Group, University of Luxembourg, Luxembourg.
Working on the development of unsupervised deepfake detection solutions under diverse environments and conditions.

Dec. 2020 – Jul. 2021 — **Student Research Assistant** – Computer Vision, Imaging and Machine Intelligence (CVI^2) Research Group, University of Luxembourg, Luxembourg.
Studied the state-of-the-art of deepfake detection and developed a supervised method for face-swap image detection using Pytorch.

Jun. 2017 – Dec. 2017 — **Intern in Computer Vision** – Etix Labs, Luxembourg.
Developed a prototype intelligent video surveillance system for intrusion detection in data centers using Python and OpenCV.

Education

Sept. 2021 - Jul. 2025 — **Ph.D. in Computer Science, University of Luxembourg**, Luxembourg.
Thesis title: *Unsupervised Anomaly Detection for Type-Agnostic and Cross-Domain Deepfake Detection*.

Sept. 2019 – Jul. 2021 — **M.Sc. Information & Computer Science, University of Luxembourg**.
Thesis title: *Face-swap Deepfake Detection Using High-frequency Components*.

Sept. 2012 – Sept. 2018 — **M.Eng. Control Systems and Automation, INSAT**, Tunisia.
Thesis title: *Research and development of computer vision algorithms for intelligent visual surveillance*.

Research Publications

Conference Proceedings

1. Nguyen, D., **Mejri, N.**, Singh, I. P., Kuleshova, P., Astrid, M., Kacem, A., … Aouada, D. (2024). Laa-net: Localized artifact attention network for quality-agnostic and generalizable deepfake detection. In *Proceedings of the ieee/cvf conference on computer vision and pattern recognition (cvpr)* (pp. 17395–17405).
2. **Mejri, N.**, Chernakov, P., Kuleshova, P., Ghorbel, E., & Aouada, D. (2024). Facial region-based ensembling for unsupervised temporal deepfake localization. In *2024 ieee international conference on multimedia and expo (icme)* (pp. 1–6). doi:10.1109/ICME57554.2024.10688329
3. **Mejri, N.**, Ghorbel, E., & Aouada, D. (2023). Untag: Learning generic features for unsupervised type-agnostic deepfake detection. In *Icassp 2023 - 2023 ieee international conference on acoustics, speech and signal processing (icassp)* (pp. 1–5). doi:10.1109/ICASSP49357.2023.10095983
4. Singh, I. P., **Mejri, N.**, Nguyen, v. D., Ghorbel, E., & Aouada, D. (2023). Multi-label deepfake classification. In *2023 ieee 25rd international workshop on multimedia signal processing (mmsp)*.

5 **Mejri, N.**, Papadopoulos, K., & Aouada, D. (2021). Leveraging high-frequency components for deepfake detection. In *2021 ieee 23rd international workshop on multimedia signal processing (mmsp)* (pp. 1–6). doi:10.1109/MMSP53017.2021.9733606

Journal Articles

1 **Mejri, N.**, Lopez-Fuentes, L., Roy, K., Chernakov, P., Ghorbel, E., & Aouada, D. (2024). Unsupervised anomaly detection in time-series: An extensive evaluation and analysis of state-of-the-art methods. *Expert Systems with Applications*, *256*, 124922. doi:https://doi.org/10.1016/j.eswa.2024.124922

2 Dupont, B., **Mejri, N.**, & Da Costa, G. (2020). Energy-aware scheduling of malleable hpc applications using a particle swarm optimised greedy algorithm. *Sustainable Computing: Informatics and Systems*, *28*, 100447. doi:https://doi.org/10.1016/j.suscom.2020.100447

Skills

Technical Skills Python, Pytorch, Unix environments

Academic Skills Research methods, LaTeX typesetting and publishing, teaching, mentoring, outreach & citizen science, volunteering in conferences.

Language Proficiency Strong reading, writing and speaking competencies in English, and French, Elementary proficiency in Luxembourgish and German.

Miscellaneous Experience

Awards and Achievements

2019 **Merit Award**, University of Luxembourg.

2020 **Merit Award**, University of Luxembourg.

2021 **Merit Award**, University of Luxembourg.

2022 **Ph.D. Industrial Fellowship grant**, National Research Fund (FNR), Luxembourg.

References

Available on Request